

Conditional probability



1 0.8

2 0.1875

3 a $\frac{2}{7}$

b $\frac{2}{9}$

4 0.4

5 a 0.42

b 0.12

c 0.88

d 0.28

e 0.6

6 a 4

b 6

c $\frac{2}{9}$

d $\frac{4}{9}$

7 a 40

b $\frac{1}{5}$

c $\frac{1}{5}$

8 a $\frac{1}{9}$

b $\frac{5}{9}$

c $\frac{1}{3}$

9 a $\frac{1}{20}$

b $\frac{2}{5}$

c $\frac{1}{5}$

d $\frac{1}{5}$

10 a $\frac{1}{6}$

b $\frac{1}{3}$

11 a $\frac{4}{13}$

b $\frac{2}{3}$

c $\frac{1}{2}$

d $\frac{4}{9}$

12 a $\frac{1}{12}$

b $\frac{1}{3}$

c $\frac{1}{4}$

13

a $\frac{9}{19}$

b $\frac{4}{19}$



c $\frac{3}{19}$

14

a 0.78

b $\frac{6}{13}$

15

a 0.0375

b 0.2375

c 0.275

d 0.14

16

a Patricia is chosen second, after Aaron has been chosen first.

b Greater than

17

a 2, 4, 6, 8, 10, 12, 14, 16, 18, 20

b 1, 2, 3, 4, 6, 8, 12, 16

c 6

d $\frac{3}{5}$

Venn diagrams

18 $\frac{1}{3}$

19

a 0.23

b 0.73

c $\frac{27}{50}$

d $\frac{23}{56}$

e $\frac{17}{50}$

20

a 6

b 2

c 5

21

a 55

b $\frac{5}{11}$

c $\frac{1}{5}$

d $\frac{2}{3}$

22

a $\frac{3}{4}$

b $\frac{5}{9}$

23

a

i 260 ii 237 iii 111 iv 208



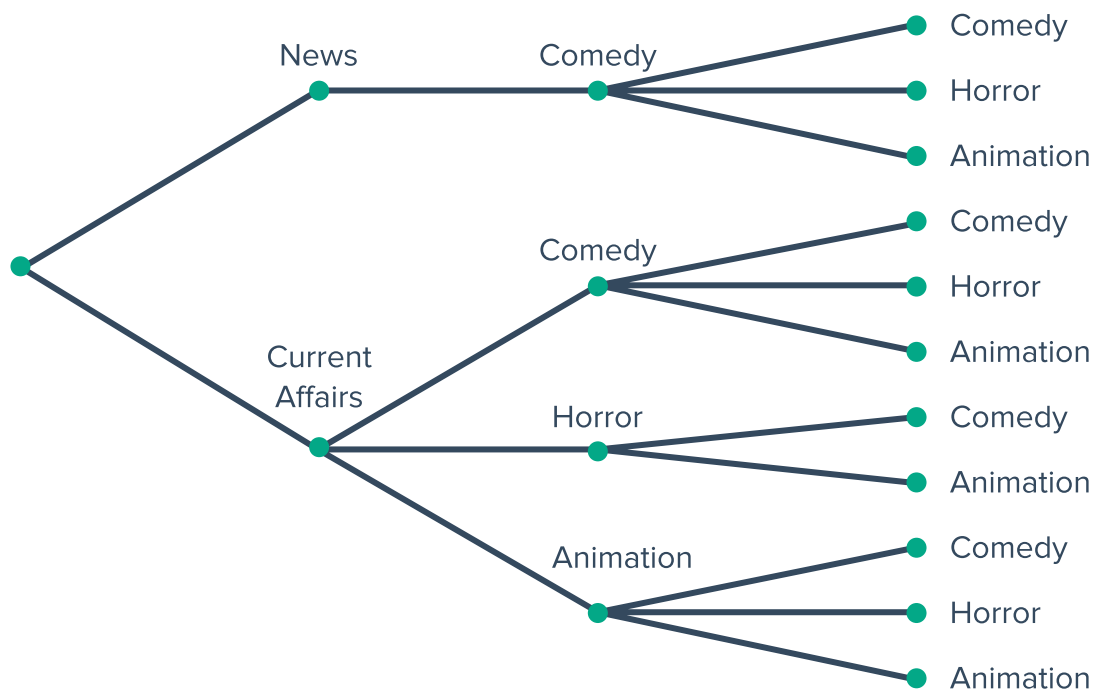
b

$$\frac{208}{319}$$

Tree diagrams and tables

24

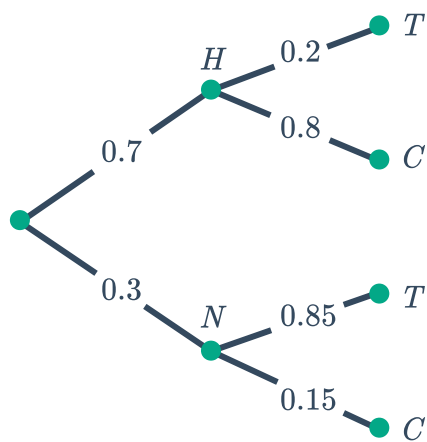
a



b $\frac{1}{3}$

25

a



b 0.35

26

a $\{0, 1, 2, 3, 4, 5\}$

b $\frac{2}{9}$



c $\frac{1}{3}$

27

a

	Read the book	Didn't read the book	Total
Watched the movie	33	52	85
Didn't watch the movie	58	8	66
Total	91	60	151

b

i $\frac{58}{151}$

ii $\frac{33}{85}$

iii $\frac{13}{15}$

28

a

	Labor (next election)	Liberal (next election)	Greens (next election)	Total
Labor (last election)	26	40	4	70
Liberal (last election)	14	75	1	90
Greens (last election)	10	25	5	40
Total	50	140	10	200

b $\frac{7}{45}$

c $\frac{1}{2}$

d $\frac{23}{32}$

29

a $\frac{5}{13}$

b $\frac{8}{13}$

c $\frac{3}{8}$

d $\frac{1}{5}$

e H and S